

2520090220

```
mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| bank_transactions |
+-----+
1 row in set (0.05 sec)

mysql> DESC bank_transactions;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| txn_id         | int           | NO   | PRI | NULL     |       |
| customer_name  | varchar(50)   | YES  |     | NULL     |       |
| branch_name    | varchar(50)   | YES  |     | NULL     |       |
| transaction_type | varchar(20)   | YES  |     | NULL     |       |
| amount         | decimal(10,2) | YES  |     | NULL     |       |
| transaction_date | date          | YES  |     | NULL     |       |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.04 sec)

mysql> INSERT INTO bank_transactions VALUES
-> (101, 'Ravi', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
-> (102, 'Sita', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
-> (103, 'Kiran', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
-> (104, 'Anil', 'Vizag', 'Deposit', 8000, '2024-01-10'),
-> (105, 'Priya', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
-> (106, 'Ramesh', 'Vizag', 'Deposit', 15000, '2024-01-12'),
-> (107, 'Keerthi', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
-> (108, 'Rahul', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
-> (109, 'Sneha', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
-> (110, 'Madhu', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
Query OK, 10 rows affected (0.05 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

```
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.22 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE bank_db;
Query OK, 1 row affected (0.01 sec)

mysql> USE bank_db;
Database changed
mysql> CREATE TABLE bank_transactions (
->   txn_id INT PRIMARY KEY,
->   customer_name VARCHAR(50),
->   branch_name VARCHAR(50),
->   transaction_type VARCHAR(20),
->   amount DECIMAL(10,2),
->   transaction_date DATE
-> );
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> GRANT SELECT
-> ON bank_transactions
-> TO Auditor1;
ERROR 1410 (42000): You are not allowed to create a user with GRANT
mysql> CREATE USER 'Auditor1'@'localhost'
-> IDENTIFIED BY 'auditor123';
ERROR 1396 (HY000): Operation CREATE USER failed for 'Auditor1'@'localhost'
mysql> SELECT User, Host
-> FROM mysql.user;
```

User	Host
Auditor1	localhost
BranchManager	localhost
mysql.infoschema	localhost
mysql.session	localhost
mysql.sys	localhost
root	localhost

6 rows in set (0.00 sec)

```
mysql> GRANT SELECT
-> ON bank_db.bank_transactions
-> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> GRANT ALL
-> ON bank_db.bank_transactions
-> TO 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> REVOKE SELECT
-> ON bank_db.bank_transactions
-> FROM 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.04 sec)
```

```
mysql> SELECT *
-> FROM bank_transactions
-> WHERE transaction_type = 'Deposit';
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Ravi	Hyderabad	Deposit	7000.00	2024-01-05
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
104	Anil	Vizag	Deposit	8000.00	2024-01-10
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16

6 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM bank_transactions
-> ORDER BY amount DESC;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
104	Anil	Vizag	Deposit	8000.00	2024-01-10
101	Ravi	Hyderabad	Deposit	7000.00	2024-01-05
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13

10 rows in set (0.00 sec)

```
mysql> USE bank_db;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_bank_db |
+-----+
| bank_transactions |
+-----+
1 row in set (0.07 sec)

mysql> SELECT * FROM bank_transactions;
+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date |
+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 7000.00 | 2024-01-05 |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.07 sec)
```

```
mysql> SELECT * FROM bank_transactions
-> WHERE txn_id = 105;
+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date |
+-----+-----+-----+-----+-----+-----+
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> DELETE FROM bank_transactions
-> WHERE txn_id = 111;
Query OK, 0 rows affected (0.04 sec)

mysql> SELECT * FROM bank_transactions;
+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date |
+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 5000.00 | 2024-01-05 |
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 |
| 103 | Kiran | Vijayawada | Deposit | 12000.00 | 2024-01-08 |
| 104 | Anil | Vizag | Deposit | 8000.00 | 2024-01-10 |
| 105 | Priya | Hyderabad | Withdrawal | 5000.00 | 2024-01-11 |
| 106 | Ramesh | Vizag | Deposit | 15000.00 | 2024-01-12 |
| 107 | Keerthi | Vijayawada | Withdrawal | 1000.00 | 2024-01-13 |
| 108 | Rahul | Hyderabad | Deposit | 9000.00 | 2024-01-14 |
| 109 | Sneha | Vizag | Withdrawal | 4000.00 | 2024-01-15 |
| 110 | Madhu | Vijayawada | Deposit | 11000.00 | 2024-01-16 |
+-----+-----+-----+-----+-----+-----+
10 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM bank_transactions;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Ravi	Hyderabad	Deposit	5000.00	2024-01-05
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06
103	Kiran	Vijayawada	Deposit	12000.00	2024-01-08
104	Anil	Vizag	Deposit	8000.00	2024-01-10
105	Priya	Hyderabad	Withdrawal	3500.00	2024-01-11
106	Ramesh	Vizag	Deposit	15000.00	2024-01-12
107	Keerthi	Vijayawada	Withdrawal	1000.00	2024-01-13
108	Rahul	Hyderabad	Deposit	9000.00	2024-01-14
109	Sneha	Vizag	Withdrawal	4000.00	2024-01-15
110	Madhu	Vijayawada	Deposit	11000.00	2024-01-16

```
10 rows in set (0.00 sec)
```

```
mysql> UPDATE bank_transactions
```

```
-> SET amount = 5000
```

```
-> WHERE txn_id = 105;
```

```
Query OK, 1 row affected (0.04 sec)
```

```
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> SELECT * FROM bank_transactions
```

```
-> WHERE txn_id = 105;
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
105	Priya	Hyderabad	Withdrawal	5000.00	2024-01-11

```
1 row in set (0.00 sec)
```

```
mysql> |
```

```
mysql>
```

```
mysql> COMMIT;
```

```
Query OK, 0 rows affected (0.04 sec)
```

```
mysql>
```

```
mysql> SELECT * FROM bank_transactions
```

```
-> WHERE txn_id IN (101,102);
```

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
101	Ravi	Hyderabad	Deposit	7000.00	2024-01-05
102	Sita	Hyderabad	Withdrawal	2000.00	2024-01-06

```
2 rows in set (0.00 sec)
```

```
mysql> |
```

```
mysql> SELECT * FROM bank_transactions
-> WHERE txn_id = 102;
+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date |
+-----+-----+-----+-----+-----+-----+
| 102 | Sita | Hyderabad | Withdrawal | 2000.00 | 2024-01-06 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> UPDATE bank_transactions
-> SET amount = 7000
-> WHERE txn_id = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>
mysql> SAVEPOINT SP1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> UPDATE bank_transactions
-> SET amount = 9000
-> WHERE txn_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql>
mysql> ROLLBACK TO SP1;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> UPDATE bank_transactions
-> SET amount = 6000
-> WHERE txn_id = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM bank_transactions
-> WHERE txn_id = 101;
+-----+-----+-----+-----+-----+-----+
| txn_id | customer_name | branch_name | transaction_type | amount | transaction_date |
+-----+-----+-----+-----+-----+-----+
| 101 | Ravi | Hyderabad | Deposit | 6000.00 | 2024-01-05 |
+-----+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql>
mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> UPDATE bank_transactions
-> SET amount = 99999
-> WHERE txn_id = 102;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1 Changed: 1 Warnings: 0
```

```
mysql> ROLLBACK TO Before_Update;
Query OK, 0 rows affected (0.04 sec)
```